Building the Capability for Innovation Synthesis

Why Complex Innovation Requires More Than Analysis, Control and Consensus

A strategic perspective for business and innovation leaders

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Executive Perspective

Organisations invest considerable effort in improving how they manage complex innovation initiatives. They introduce governance structures, programme management methodologies, performance dashboards and increasingly sophisticated information systems. The underlying assumption is that better analysis, planning and control will lead to more predictable outcomes.

Yet complex innovation programmes frequently encounter difficulties that cannot be explained by inadequate planning or execution alone.

Different stakeholders interpret the challenge differently. Functional experts develop solutions that make sense within their own domains but prove difficult to integrate. Management teams struggle to reconcile competing priorities. As uncertainty increases, the natural response is often to introduce more detailed plans, additional reporting and tighter controls.

These interventions may improve visibility without resolving the underlying problem.

The difficulty is that complex innovation cannot always be managed as the implementation of a solution that has already been defined. When technological, commercial and organisational variables are highly interconnected, the appropriate solution may only emerge through interaction, experimentation and collective learning.

This creates a fundamental leadership challenge.

How can organisations develop solutions when no individual function or management team possesses all the necessary knowledge, and when stakeholders hold different, sometimes opposing, perspectives?

The answer requires more than analytical capability and programme discipline. It requires the ability to create shared understanding, build collective ownership and develop solutions through collaboration.

> **Innovation synthesis is not the outcome of a single creative workshop. It is an organisational capability that develops through shared understanding, collective ownership and continuous adaptation.**

This whitepaper explores how organisations can develop that capability and introduces the Innovation Synthesis Maturity Model as a framework for understanding their progress.

1. Why Traditional Programme Management Falls Short

Management approaches have evolved considerably over recent decades.

For much of the twentieth century, strategic decisions relied heavily on the experience and judgement of senior executives. As organisations grew, management models provided more structured ways to analyse markets, organisational capabilities and investment choices.

The development of information technology subsequently made increasingly detailed operational information available. Enterprise Resource Planning systems enabled organisations to integrate data across functions, while dashboards and reporting systems strengthened management's ability to monitor performance.

These developments significantly improved organisational coordination and control.

They also reinforced an assumption that continues to influence complex transformation programmes: if an organisation can analyse the situation thoroughly, design an appropriate solution and control its implementation, the intended outcome should follow.

For relatively predictable activities, this logic remains valuable.

For complex innovation challenges, however, it can become problematic.

The limitations of functional decomposition

Traditional programme management commonly operates through two interconnected dimensions.

The first is the functional or substantive dimension. A complex challenge is divided into separate areas of expertise. Each function develops its contribution to the overall solution.

The second is the programme management dimension, often characterised by Command & Control. Responsibilities, milestones, budgets and reporting structures are established to coordinate implementation and monitor progress.

Both dimensions are necessary.

Functional expertise provides specialised knowledge. Programme management creates coordination, accountability and transparency.

The difficulty arises when the overall challenge cannot be understood simply by combining independently developed functional solutions.

A technological decision may influence customer propositions, operational processes, organisational responsibilities and employee capabilities simultaneously. Each function may optimise its own contribution while the organisation struggles to develop a coherent overall solution.

Integration then becomes more than a technical coordination exercise.

It becomes a process of reconciling different interpretations of the problem itself.

When more control reinforces the problem

As implementation difficulties emerge, programme managers often respond by increasing control.

Reporting becomes more detailed. Deadlines are reinforced. Deviations require additional justification. Functional teams are encouraged to concentrate on delivering their assigned responsibilities.

These responses may be entirely reasonable when the solution is sound and execution discipline is lacking.

But when the underlying problem has been misunderstood, additional control can reinforce the original assumptions rather than challenge them.

Functional experts may become increasingly focused on protecting their own deliverables. Reporting can become defensive. Emerging doubts about the overall solution remain unspoken because acknowledging them may be interpreted as poor performance.

The programme gradually loses its ability to learn.

The paradox is that measures intended to improve delivery can reduce the openness needed to recognise that the chosen solution requires reconsideration.

> **When the challenge is complex, stronger control cannot compensate for the absence of shared understanding.**

This does not mean abandoning programme management.

It means recognising that functional expertise and Command & Control, although necessary, are insufficient when the solution itself remains uncertain.

2. The Missing Dimension: Connect & Commit

Complex innovation challenges rarely fail because organisations lack intelligent people.

They struggle because available knowledge, experience and perspectives are not effectively brought together.

Different stakeholders may understand individual aspects of the challenge, yet no shared interpretation emerges. Decisions are made without sufficient understanding of their implications for other functions. Resistance develops because people do not recognise their own concerns or assumptions in the chosen direction.

Bart Roestenberg addresses this problem in his work on The Third Dimension of Programme Management.

He argues that traditional programme management needs a third dimension alongside functional expertise and Command & Control.

He calls this dimension Connect & Commit.

Connect: developing shared understanding

Connecting stakeholders involves more than establishing communication channels or organising regular meetings.

Formal communication can ensure that information reaches the right people without creating genuine understanding.

Connect operates at a deeper level.

It requires stakeholders to explore how they interpret the challenge, what they consider important and which assumptions influence their decisions.

This is particularly relevant when innovation involves different professional cultures.

Engineers, commercial managers, financial specialists and operational leaders may use the same terminology while attaching different meanings to success, risk, quality or value.

These differences often remain invisible until they produce conflict.

Creating connection means making those perspectives discussable.

It involves recognising that disagreement may originate not from unwillingness to cooperate but from different interpretations of organisational reality.

This connects directly to the discussion of organisational culture in Whitepaper 02, Leading Innovation in Complexity.

Formal structures influence behaviour, but so do values, beliefs and deeply embedded assumptions.

An effective Connect process creates the conditions in which these underlying perspectives can be explored constructively.

Commit: creating collective ownership

Understanding different perspectives is necessary, but it does not automatically lead to commitment.

People may understand a decision without feeling responsible for its success.

Commit therefore goes beyond communicating objectives or obtaining formal approval.

It concerns the development of intrinsic motivation and collective responsibility.

Stakeholders need to understand not only what the programme intends to achieve, but also why the outcome matters and how it affects their responsibilities, interests and contributions.

When people participate meaningfully in developing a solution, they are more likely to recognise their own contribution to its success.

This does not imply that every participant must agree with every decision.

It means that the decision-making process creates sufficient understanding, trust and ownership for stakeholders to contribute constructively to implementation.

A third dimension, not an alternative management philosophy

Connect & Commit should not be positioned as a replacement for functional expertise or Command & Control.

Complex innovation requires all three.

Functional expertise provides the substantive knowledge needed to understand the challenge.

Command & Control provides coordination, decision-making discipline and accountability.

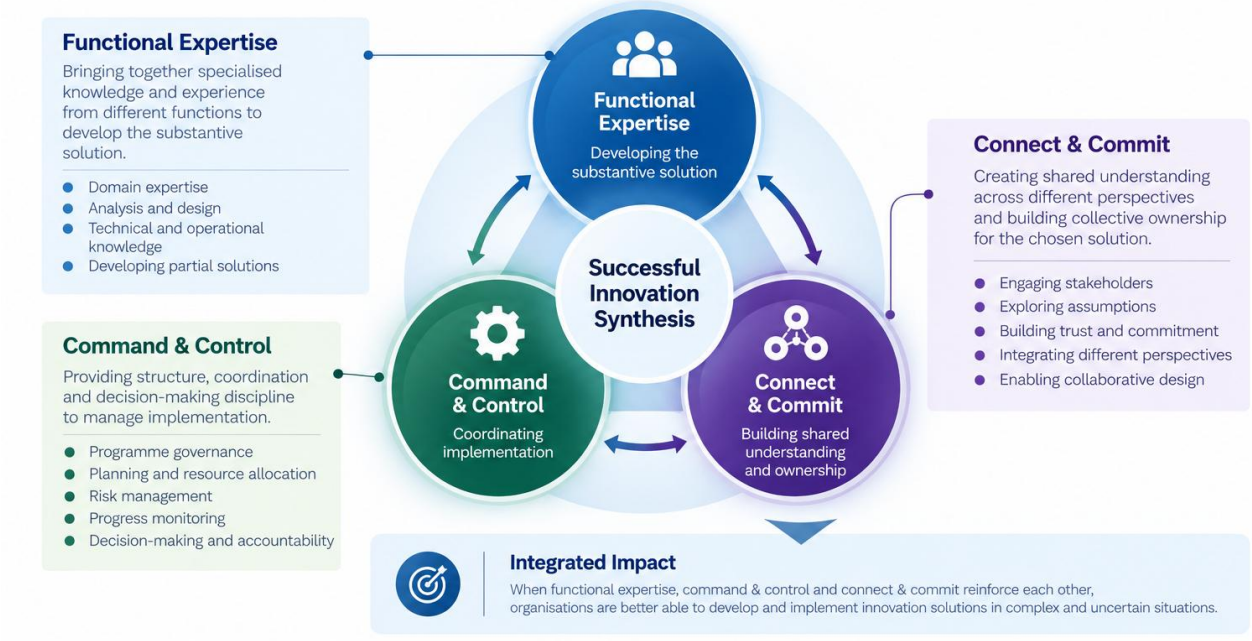
Connect & Commit enables stakeholders to integrate their perspectives and develop the shared understanding and ownership required for implementation.

FIGURE 1

The Three Dimensions of Complex Innovation Programmes

Technical expertise, programme management and human connection work together to develop and implement innovation solutions.

Complex innovation programmes require more than analytical capability and control. They also need a deliberate focus on connecting people and creating collective ownership. The three dimensions reinforce each other and are all necessary for successful innovation synthesis.



The critical point is that Connect & Commit cannot be treated as an occasional communication activity.

It must be deliberately designed, resourced and managed.

Roestenberg argues that responsibility for this dimension should be explicitly assigned within the programme, supported by the authority and resources required to make it effective.

Without this structural commitment, Connect & Commit risks becoming a collection of well-intentioned workshops rather than an integral part of programme governance.

3. Designing the Conditions for Innovation Synthesis

Recognising the importance of shared understanding is relatively straightforward.

Creating the conditions in which it can emerge is considerably more difficult.

Organisations are accustomed to meetings in which information is presented, positions are defended and decisions are requested.

These formats are not necessarily suitable when the objective is to develop a new understanding of a complex challenge.

The American innovators Matt and Gail Taylor explored how groups with different backgrounds and perspectives could work together creatively.

Their approach starts from an important assumption: diversity of knowledge and experience is not an obstacle to collective problem-solving.

It is a potential source of better solutions.

However, this potential is realised only when the collaborative process is deliberately designed.

Four conditions for collective creativity

The Taylors identified four important conditions for effective collaborative creative processes.

The right people

The process must involve stakeholders representing the relevant perspectives, knowledge and experience.

This extends beyond formal decision-makers and designated subject experts. People who understand operational realities, customer experiences or organisational consequences may contribute insights that are otherwise overlooked.

The right content

Participants need access to relevant information, evidence and expertise.

Some knowledge is explicit and can be shared through data, models, presentations and visualisations. Other knowledge is tacit and emerges through the experiences and interpretations of participants.

An effective process brings these forms of knowledge together.

The right process

Complex problems require more than an open discussion.

A deliberately designed process helps participants explore assumptions, develop shared insights, experiment with possibilities and translate emerging ideas into practical solutions.

The process must be sufficiently structured to maintain direction while allowing unexpected insights to emerge.

The right environment

The physical environment can influence how people interact and think.

An open, creative space allows participants to work in different group configurations, exchange ideas and make their thinking visible.

Visualising information and emerging conclusions helps participants develop a shared understanding that extends beyond individual conversations.

Together, these four conditions create an environment in which different perspectives can contribute to collective problem-solving.

From dialogue to collaborative design

Building on these principles, GroupMapping uses the Creative Intervention Model to structure collaborative interventions.

The model consists of four steps.

Share

Participants share perspectives, experiences, assumptions and objectives.

The purpose is to establish a broader understanding of the challenge and recognise differences that may otherwise remain hidden.

Learn

Participants collectively explore the relevant knowledge, evidence and implications.

This creates a more informed foundation for developing potential solutions.

Play

Participants experiment with alternative interpretations and possibilities.

Creative exploration makes it possible to challenge established assumptions without immediately committing to a particular solution.

Design

Emerging insights are translated into feasible solution directions and practical arrangements.

The objective is to develop an outcome that stakeholders understand, have helped shape and are prepared to support.

The four steps provide a meta-agenda rather than a rigid sequence of workshop activities.

Their value lies in creating a structured progression from different individual perspectives towards shared understanding and collaborative design.

Why a workshop is not enough

Creative interventions can be powerful instruments for developing Connect & Commit.

However, their effectiveness depends on how they are connected to the broader innovation programme.

A workshop may generate enthusiasm and promising ideas, but these can quickly disappear when participants return to organisational structures and performance systems that reinforce established behaviour.

For this reason, the timing, purpose and follow-up of interventions are critical.

The objective is not to organise as many collaborative events as possible.

It is to create the necessary moments for collective learning and decision-making, and ensure that their outcomes influence the programme.

Not every project requires an extensive Connect & Commit approach.

Where objectives are clear, uncertainty is limited and stakeholders broadly agree on the solution, conventional project management may be sufficient.

The additional investment becomes particularly valuable when the organisation faces genuinely complex innovation challenges involving conflicting perspectives and uncertain outcomes.

4. The Innovation Synthesis Maturity Model

Developing an innovation synthesis requires capabilities that many organisations have not systematically developed.

Recognising competing perspectives is only the beginning.

Organisations must also learn to analyse the underlying paradoxes, understand their implications for value creation, develop shared solutions and adapt those solutions as circumstances change.

The Innovation Synthesis Maturity Model, developed by Klokgieters, describes four levels of organisational development.

Each level represents a progression in the organisation's ability to address complex innovation challenges through paradox thinking and collaborative synthesis.

Level 1: I-Paradox Awareness

At the first level, organisations begin to recognise that complex innovation challenges may involve legitimate but apparently contradictory perspectives.

Previously, disagreements may have been resolved primarily through hierarchical authority, political influence or repeated attempts to demonstrate that one interpretation was correct.

The central challenge is to create awareness that different perspectives may both contain valuable insights.

Stakeholders must become willing to examine their own assumptions and understand those of others.

This requires the ability to identify relevant innovation paradoxes and establish conditions for constructive dialogue.

The twelve innovation paradoxes introduced in Whitepaper 03 provide a useful starting point.

They offer a common language for identifying tensions that may otherwise remain hidden within recurring management discussions.

The organisation's emerging capability is therefore not yet the development of a synthesis.

It is the ability to recognise when paradox thinking may be relevant and to engage stakeholders in exploring their different perspectives.

Level 2: I-Paradox Analysis

Awareness alone does not provide sufficient direction for strategic decision-making.

An organisation may recognise several innovation paradoxes without understanding which have the greatest implications for value creation.

At the second level, the organisation develops the capability to analyse these tensions systematically.

This involves identifying the relevant stakeholders, understanding the assumptions behind their perspectives and examining the potential consequences of different strategic choices.

Cross-functional knowledge becomes essential.

The organisation may also need external expertise where internal knowledge is insufficient.

Scenario thinking plays an important role because complex innovation challenges rarely allow the consequences of alternative choices to be predicted with certainty.

Different scenarios can make underlying assumptions explicit and help stakeholders explore how contrasting perspectives influence potential outcomes.

The objective is to identify the innovation paradoxes with the greatest strategic impact and determine where collaborative solution development should be concentrated.

The organisation moves from recognising tensions to understanding their implications.

Level 3: Static I-Synthesis

At the third level, the organisation develops the capability to formulate and implement a shared innovation synthesis.

Stakeholders no longer limit themselves to analysing competing perspectives.

They collaborate in developing a solution that combines relevant insights and addresses the underlying innovation challenge.

The process requires shared understanding, creative exploration and collective ownership.

The organisation develops a recognisable approach, supported by appropriate methods and instruments, for bringing stakeholders together and designing solutions.

Connect & Commit becomes particularly important because the synthesis must be supported by those responsible for implementing it.

The defining characteristic of this level is the ability to move from analysis to a collectively developed solution.

However, the synthesis is still treated largely as a defined outcome.

Once the solution has been selected and implemented, the organisation may assume that the original paradox has been adequately addressed.

This assumption becomes the central limitation of Static I-Synthesis.

Level 4: Dynamic I-Synthesis

At the highest maturity level, the organisation recognises that innovation syntheses are not necessarily permanent solutions.

Technologies evolve, markets change and organisational capabilities develop.

A synthesis that was effective under previous circumstances may become less appropriate as new developments emerge.

The organisation therefore develops the ability to reconsider and adapt previously established solutions.

This requires continuous attention to relevant innovation paradoxes, emerging knowledge and changing conditions.

Connect & Commit is no longer dependent on individual initiatives or occasional interventions.

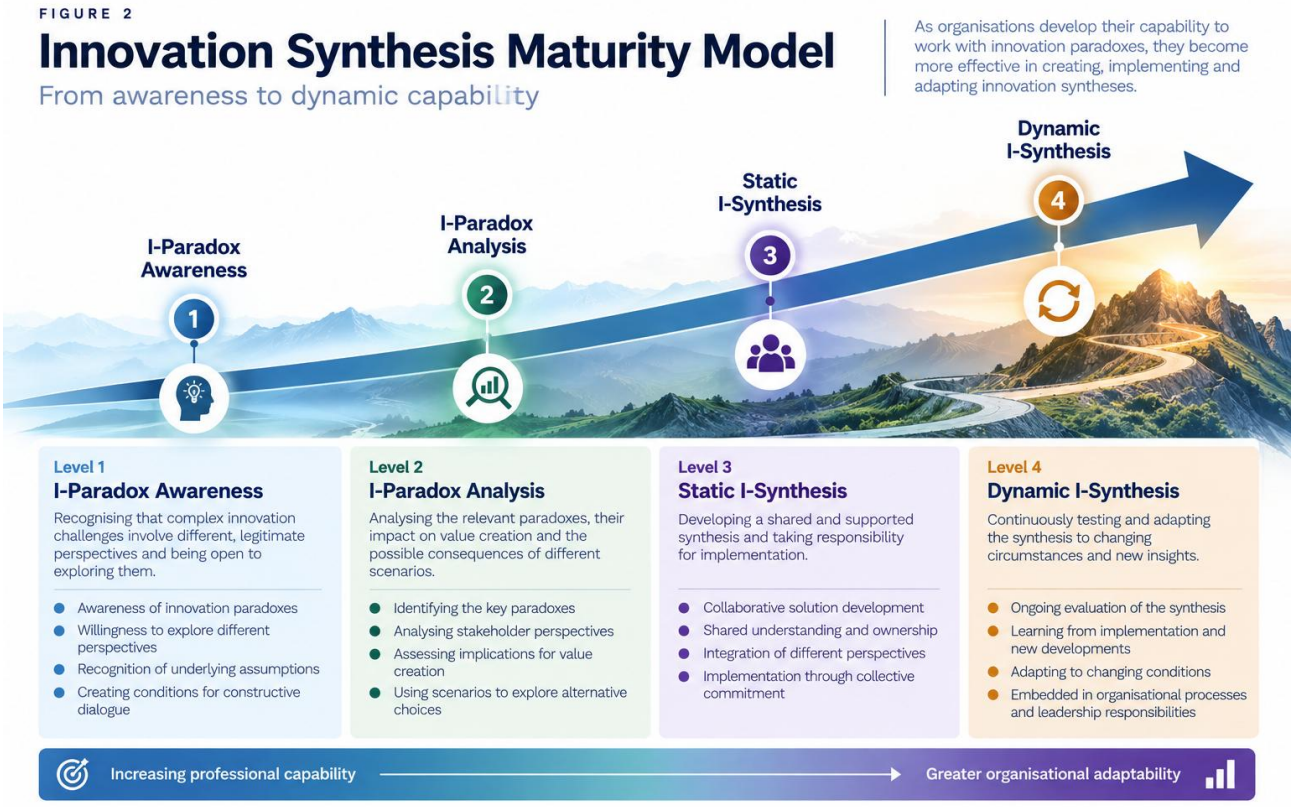
It becomes structurally embedded in the organisation's innovation processes and leadership responsibilities.

Internal and external knowledge are brought together through ongoing collaboration.

The organisation may develop permanent arrangements, including creative environments and innovation ecosystems, that support continuous learning and collective problem-solving.

The distinguishing capability is organisational adaptability.

Rather than repeatedly treating complex innovation challenges as isolated problems, the organisation becomes increasingly capable of recognising tensions, developing syntheses and revising them as circumstances change.



Maturity is about capability, not methodology

The four levels should not be interpreted as a checklist of management instruments.

An organisation may use creative workshops, scenario analysis and sophisticated programme management methods without having developed the underlying capabilities associated with higher maturity.

The decisive question is whether these approaches have changed how the organisation recognises complex challenges, integrates perspectives and develops solutions.

The model therefore provides a framework for reflecting on organisational capability rather than simply assessing compliance with a prescribed process.

5. From Occasional Intervention to Organisational Capability

Many organisations recognise the importance of collaboration and stakeholder engagement.

Yet they continue to approach these activities as supporting elements of programme management rather than capabilities requiring deliberate development.

Communication plans are established. Stakeholder meetings are scheduled. Workshops are organised at important milestones.

These activities can be useful, but their existence does not demonstrate that Connect & Commit has become an integral part of the organisation's innovation capability.

The more fundamental question is whether the organisation has developed a reliable way to bring different perspectives together when complex decisions must be made.

Designing responsibility for collaboration

When Connect & Commit is considered important but remains everyone's responsibility, it may become no one's priority.

Traditional programme governance assigns clear responsibility for budgets, planning, deliverables and technical integration.

The development of shared understanding and collective ownership deserves comparable attention in genuinely complex programmes.

This means establishing responsibility for designing and facilitating the necessary interactions between stakeholders.

It also means ensuring that the outcomes of those interactions influence substantive decisions and programme priorities.

Without this connection, collaborative interventions risk becoming disconnected from the mechanisms through which the organisation actually operates.

Creating space for constructive disagreement

Innovation synthesis requires stakeholders to challenge assumptions and explore perspectives that may conflict with established organisational interests.

This is difficult in environments where disagreement is interpreted as resistance or where raising uncertainty is considered a weakness.

Leaders must create conditions in which different perspectives can be expressed without immediately becoming contests for influence.

Such openness does not remove the need for accountability.

It improves the quality of accountability by making assumptions, risks and alternative interpretations more visible.

The purpose is not endless discussion.

It is to develop sufficient shared understanding to make better decisions and strengthen commitment to their implementation.

Embedding learning and adaptation

The transition from Static to Dynamic I-Synthesis is particularly significant.

It requires organisations to reconsider their understanding of implementation.

A successful implementation is not necessarily the end of the innovation challenge.

It may be the beginning of a new period in which the effectiveness of the chosen synthesis must be observed and reassessed.

This requires feedback from operations, customers, partners and other stakeholders.

It also requires the willingness to revisit decisions when new evidence challenges earlier assumptions.

Dynamic I-Synthesis therefore connects collaborative innovation with organisational learning and adaptive leadership.

The organisation becomes less dependent on occasional interventions and more capable of responding to complex innovation challenges as a normal part of its operations.

> **The ultimate objective is not to make every innovation programme more collaborative. It is to develop the organisational judgement and capability to recognise when collective synthesis is necessary and how to achieve it effectively.**

Leadership Reflection

1. Programme management

When complex innovation programmes encounter difficulties, do we primarily increase control, or do we reconsider whether stakeholders share a sufficient understanding of the challenge?

2. Connect & Commit

Is the development of shared understanding and collective ownership explicitly organised, with clear responsibilities and sufficient resources?

3. Collaborative problem-solving

Do we deliberately bring together the right people, knowledge, processes and environments when addressing complex innovation challenges?

4. Innovation Synthesis Maturity

Which of the four maturity levels best describes our current organisational capability? What prevents us from progressing to the next level?

5. Continuous adaptation

Do we treat an implemented innovation synthesis as a completed solution, or do we regularly reconsider its effectiveness as circumstances change?

These questions are intended to stimulate reflection on the organisation's ability to navigate complexity, rather than simply evaluate the performance of individual programmes.

Closing Perspective

Complex innovation challenges cannot always be resolved through better analysis, more detailed planning or stronger programme control.

These capabilities remain important.

But when stakeholders interpret the same challenge differently, when organisational consequences are uncertain and when no single function possesses all the necessary knowledge, a different capability becomes essential.

The ability to develop a shared understanding of the challenge and collaboratively design a solution.

Connect & Commit provides the missing dimension that enables functional expertise and programme management to operate more effectively under these conditions.

Creative interventions can help establish this connection, but lasting impact requires more than occasional workshops.

Organisations must develop the capability to recognise innovation paradoxes, analyse their implications, formulate shared syntheses and adapt those syntheses as circumstances change.

The Innovation Synthesis Maturity Model describes this progression.

At its highest level, innovation synthesis is no longer an exceptional activity reserved for difficult programmes.

It becomes part of how the organisation learns, makes decisions and adapts.

> **The future of innovation management will not be determined solely by how effectively organisations control complex programmes, but by how successfully they develop the collective capability to navigate complexity together.**

About the Author

Koen Klokgieters is the founder of InnoPlot and has spent more than three decades working at the intersection of strategy, innovation, technology and organisational transformation. His experience combines corporate leadership, consulting, entrepreneurship and education.

Through InnoPlot, he supports organisations and their leaders in translating strategic challenges into innovation capabilities, organisational change and measurable impact.

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Conceptual Foundations

This whitepaper is adapted from Working on Innovation Synthesis by Koen Klokgieters.

The approach draws on the strategic paradox perspective developed by Bob de Wit and Ron Meyer, the collaborative creativity principles developed by Matt and Gail Taylor, and Bart Roestenberg's Third Dimension of Programme Management, including Connect & Commit.

The Creative Intervention Model, with its four steps Share, Learn, Play and Design, is attributed to GroupMapping.

The Innovation Synthesis Maturity Model and its four maturity levels are retained from the original framework developed by Klokgieters.

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